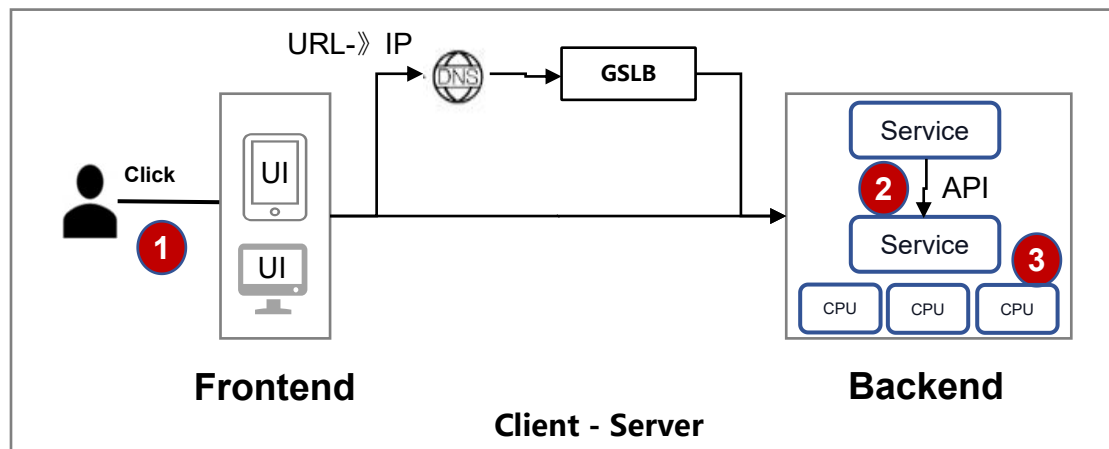


Internet of Agent and DIN

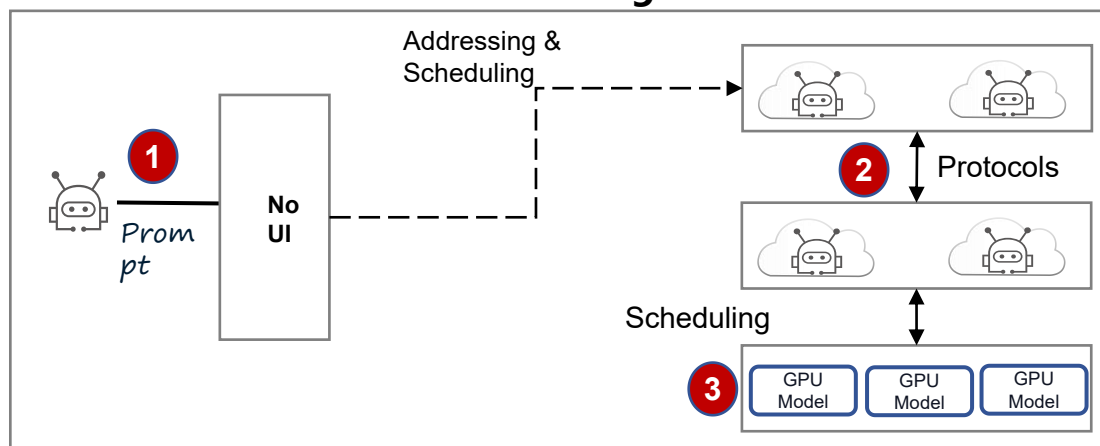
- Opportunities and Challenges

New Changes and Key Challenges

Traditional Architecture



Internet of Agent



Changes

- 1** The way to initiate request: Click with UI vs. No UI
- 2** Communications between services: API vs. Protocols
- 3** Compute Methods: CPU + SW vs. GPU + LLM

Key Challenges

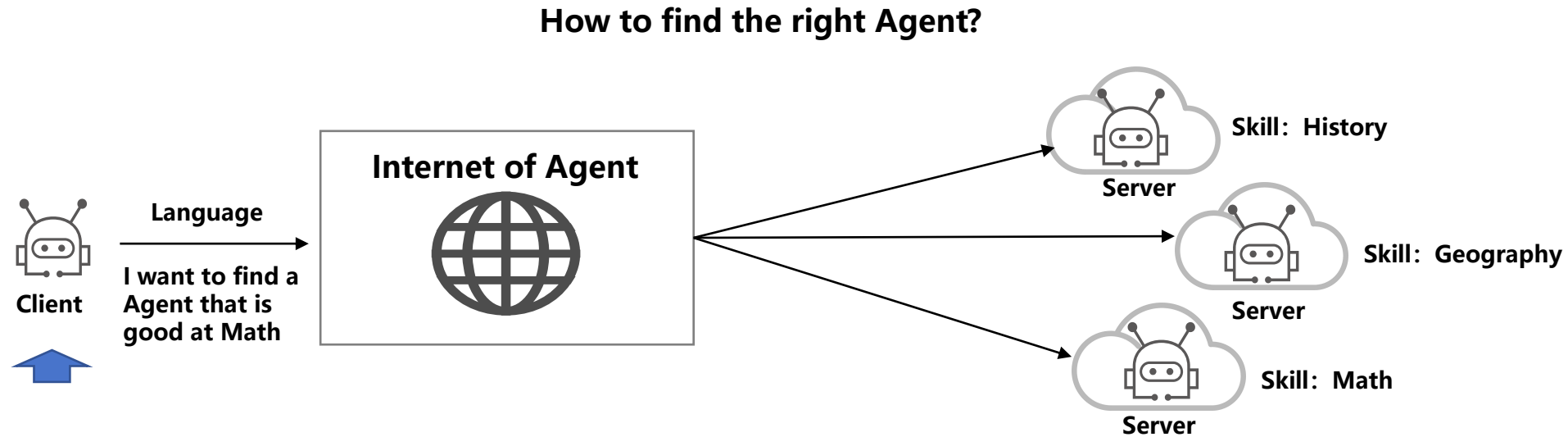
Addressing: How to find the Agents?

Protocols: How to communicate with Agents?

Scheduling: How to schedule the tasks and assign to Agents?

Security: How to protect privacy and ensure secure communications?

Addressing system and protocols for the internet of agent



The current typical process is to feed the Client with context and task, and then the Client will select the appropriate tools and agents.

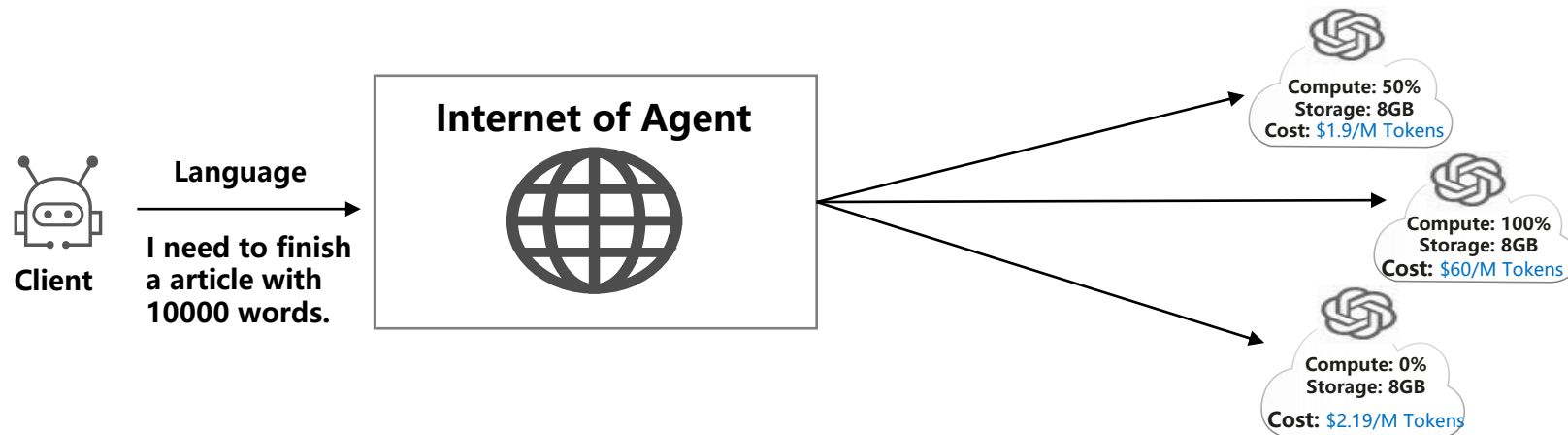
There are some challenges in this way,

1. The context window size is very limited, and cannot hold all the input context and task
2. Various Clients have various capabilities of compute resources and models, so the capability of selecting the right agent will be different
3. When the tools and agents are updated, the models in the Client will need to be retrained.

Requirements: A device/gateway needs to be deployed to collect and synchronize Agents related information, and find the right agent according to the language.

Scheduling system and protocols for the internet of agent

How to find the right server and model?

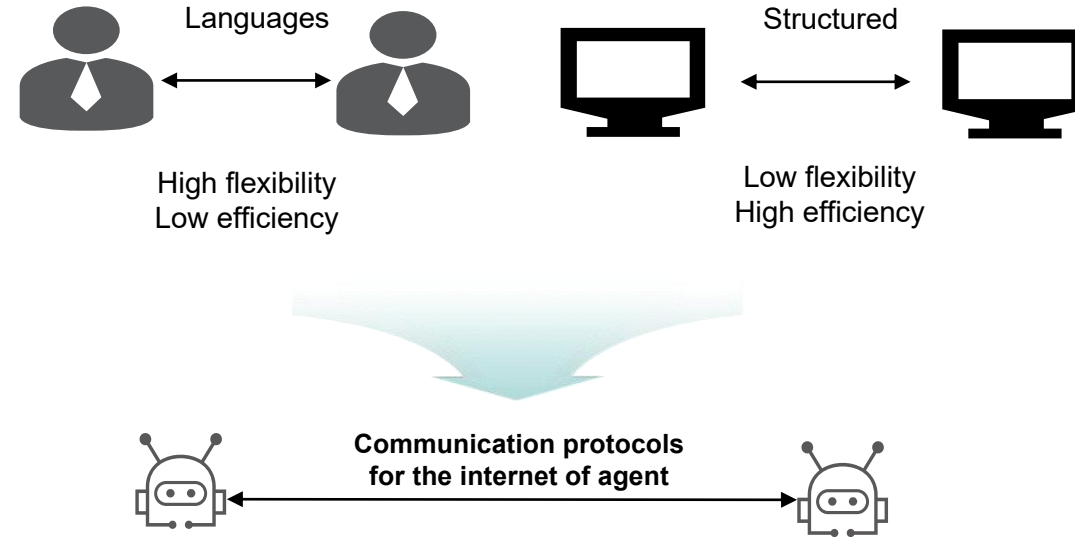


Different servers and models have different capabilities for dealing with different tasks.

Requirements: A device needs to be deployed to monitor the capabilities of the servers and models as well as the compute and storage resources.

Communication protocol for the internet of agent

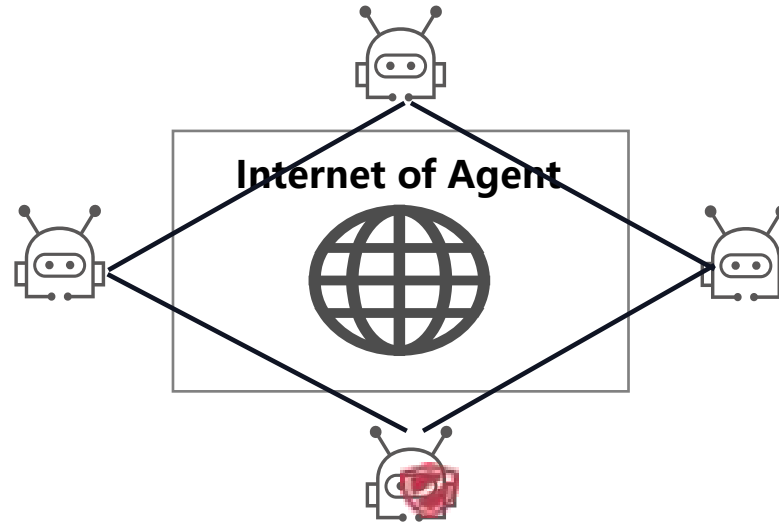
How do agents communicate with each other?



The human language has very high flexibility but very low efficiency.
The Machine language has very low flexibility but very high efficiency.

Requirements: Communication protocol for the internet of agent needs to be highly flexible and highly efficient.

Security mechanisms for the internet of agent



The boundary of the multi-agent system becomes blurred.

Security challenges for the internet of agent:

1. Content detection
2. Data protection
3. Self-security

Requirements: Security mechanisms are needed for Agent authentication, access control and privacy protection.

The evolution of the internet of agents

1st Phase~2025

Apps to Agents

2nd Phase~2027

Multi-Agent System

3rd Phase~2030

Internet of Agents

The Challenges

1. **Addressing:** How to find the Agents?
2. **Protocols:** How to communicate with Agents?
3. **Scheduling:** How to schedule the tasks and assign to Agents?
4. **Security:** How to protect privacy and ensure secure communications?

The Requirements on Networks

1. A device/gateway needs to be deployed to **collect and synchronize Agents related information**, and find the right agent according to the language.
2. A device needs to be deployed to **monitor the capabilities** of the servers and models as well as the compute and storage resources.
3. **Communication protocol** for the internet of agent needs to be highly flexible and highly efficient.
4. **Security mechanisms** are needed for Agent authentication, access control and privacy protection.

